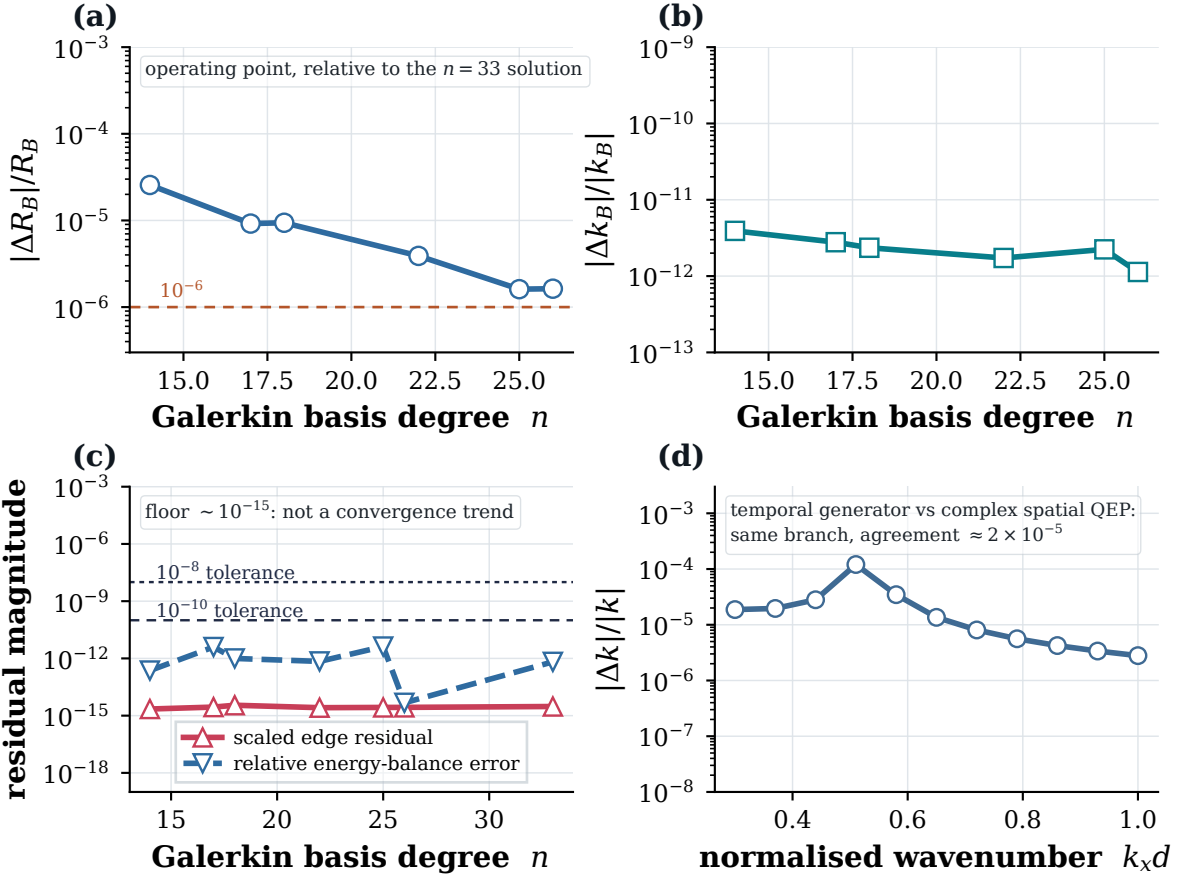
(a)
operating point, relative to the $n = 33$ solution
$|\Delta R_B|/R_B$
10^{-3}
10^{-4}
10^{-5}
10^{-6}
10^{-6}
15.0
17.5
20.0
22.5
25.0
Galerkin basis degree n
(b)
$|\Delta k_B|/|k_B|$
10^{-9}
10^{-10}
10^{-11}
10^{-12}
10^{-13}
15.0
17.5
20.0
22.5
25.0
Galerkin basis degree n
(c)
residual magnitude
floor $\sim 10^{-15}$: not a convergence trend
10^{-8} tolerance
10^{-10} tolerance
10^{-3}
10^{-6}
10^{-9}
10^{-12}
10^{-15}
10^{-18}
scaled edge residual
relative energy-balance error
15
20
25
30
Galerkin basis degree n
(d)
temporal generator vs complex spatial QEP:
same branch, agreement $\approx 2 \times 10^{-5}$
$|\Delta k|/|k|$
10^{-3}
10^{-4}
10^{-5}
10^{-6}
10^{-7}
10^{-8}
0.4
0.6
0.8
1.0
normalised wavenumber $k_x d$